Acoustic Structure of Green-cheeked Conure Vocalizations: An Unsupervised Machine Learning Approach

Portfolio Project — Master's Application
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Species: Pyrrhura molinae (Green-cheeked Parakeet)
Methods: Bioacoustics | Signal Processing | K-Means Clustering | PCA

Abstract

This project presents an exploratory acoustic analysis of Pyrrhura molinae (green-cheeked conure) vocalizations using unsupervised machine learning. Despite being a widely kept companion parrot, P. molinae remains critically understudied in the bioacoustics literature, with no published machine learning-based vocalization classification to date. Using 27 high-quality field recordings sourced from the Xeno-canto citizen science database, 254 fixed-length audio segments were extracted and characterized using a 34-dimensional acoustic feature vector comprising Mel-frequency cepstral coefficients (MFCCs), chroma features, spectral centroid, and spectral contrast. K-Means clustering identified six acoustically distinct vocalization categories, validated through silhouette analysis (score: 0.2595) and visualized via principal component analysis (PCA). Results reveal that conure vocalizations exhibit detectable but overlapping acoustic structure, consistent with a continuous communicative repertoire rather than discrete categorical calls. One cluster demonstrated clear spatial isolation in PCA space, suggesting a structurally unique vocalization type warranting further behavioral investigation. This work establishes a foundational acoustic taxonomy for P. molinae and proposes a framework for future behavioral annotation studies.

1. Introduction and Motivation

The question of whether non-human animals possess structured communicative systems has fascinated researchers for decades. Landmark studies on bottlenose dolphins have demonstrated that specific vocalizations recur in highly predictable behavioral contexts — for example, a consistent acoustic signal exchanged between dolphins immediately before coordinated task execution, analogous to a readiness signal. Similarly, research on great apes has revealed systematic associations between call types and social or environmental stimuli. These findings suggest that acoustic structure, even without full semantic mapping, can carry meaningful communicative information.

Psittacines — parrots — represent a uniquely compelling subject for such investigation. They are among the few non-human taxa capable of vocal learning, producing calls that are modified through social experience rather than fixed by genetics alone. The green-cheeked conure (Pyrrhura molinae), native to the forests of South America, is one of the most commonly kept companion parrots globally. Yet despite their behavioral richness and vocal complexity, P. molinae vocalizations have received virtually no systematic acoustic or computational study. This project is motivated by that gap.

The author's personal experience keeping and observing green-cheeked conures — including direct behavioral observation of vocalization patterns in social and solitary contexts — informed the design of this study and provided qualitative hypotheses that guided interpretation of clustering results. This project does not claim to decode conure communication. Rather, it applies rigorous signal processing and unsupervised machine learning to establish the first acoustic taxonomy of P. molinae vocalizations, laying the groundwork for future behavioral correlation studies.

2. Research Gap

The bioacoustics literature on psittacine vocalizations is heavily concentrated on a small number of species: budgerigars (Melopsittacus undulatus), orange-fronted parakeets (Eupsittula canicularis), and several large parrot species. Zebra finches and corvids have received extensive computational treatment. By contrast, Pyrrhura molinae appears in no published machine learning-based acoustic classification study. Existing descriptions of P. molinae vocalizations are limited to informal naturalist accounts and brief notes in avicultural literature.

This absence is scientifically significant. A validated acoustic feature space for P. molinae is a prerequisite for any downstream behavioral annotation, emotion-call correlation, or inter-individual communication study. The present work fills this gap at the foundational level.

3. Methodology

3.1 Data Collection

Audio recordings were sourced from Xeno-canto (xeno-canto.org), an open-access citizen science database of wildlife sound recordings. To ensure acoustic quality, only recordings assigned a quality grade of A (highest) by the Xeno-canto community were included. This filtering reduced the dataset to 27 recordings, eliminating files with excessive background noise, low signal-to-noise ratio, or poor microphone quality. Files included both wild and captive recordings, representing natural flock behavior, flight calls, contact calls, and undescribed vocalizations.

3.2 Audio Segmentation

Each recording was divided into fixed-length segments of 3.0 seconds using a sliding window approach. Trailing segments shorter than 70% of the window length were discarded to avoid feature distortion from incomplete segments. This process yielded 254 segments across the 27 source files, providing sufficient granularity for intra-file vocalization variation while maintaining computational tractability.

3.3 Feature Extraction

Each 3-second segment was represented as a 34-dimensional acoustic feature vector using the librosa Python library. Features were selected to capture complementary dimensions of vocal structure:

- Mel-frequency cepstral coefficients (MFCCs, 20 coefficients): capture timbral texture and spectral envelope shape — the primary acoustic 'fingerprint' of a sound.
- Chroma features (12 values): represent pitch class energy distribution, distinguishing tonal from noisy vocalizations.
- Spectral centroid (1 value): encodes perceived brightness — high values indicate sharp, high-frequency calls.
- Spectral contrast (1 value): measures peak-to-valley energy difference, distinguishing clear calls from diffuse background noise.

All features were standardized using scikit-learn's StandardScaler prior to clustering, ensuring equal contribution from features with different numerical ranges.

3.4 Clustering and Validation

K-Means clustering was applied to the scaled feature matrix. The optimal number of clusters k was determined through two complementary methods: the Elbow Method (minimizing within-cluster inertia) and silhouette analysis (maximizing mean silhouette coefficient). Both methods were evaluated across $k = 2$ to $k = 9$. $k = 6$ was selected as the optimal value, balancing statistical validity (silhouette score: 0.2595) with interpretability given the dataset size. Dimensionality reduction via Principal Component Analysis (PCA) to two components was used for cluster visualization.

4. Results

4.1 Cluster Selection

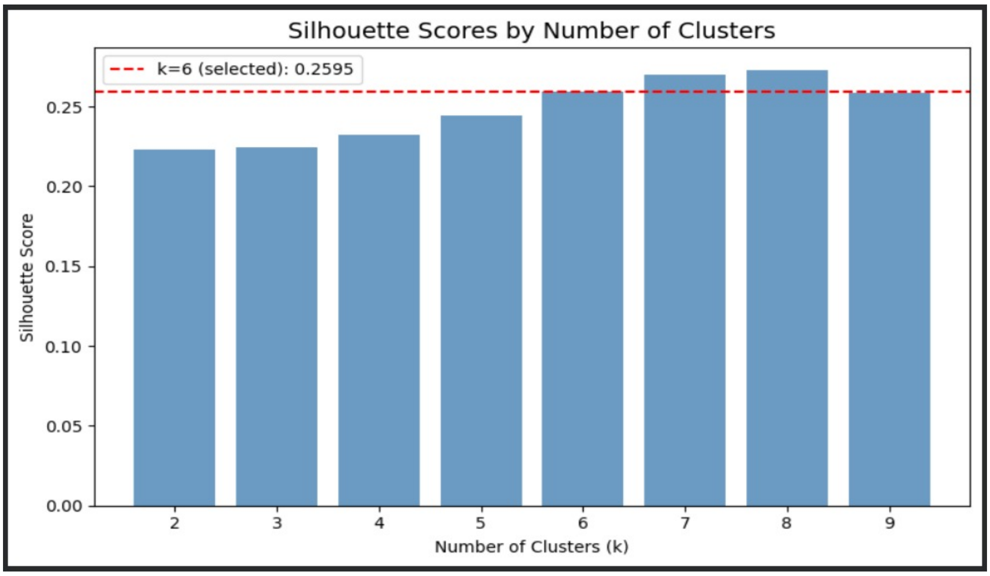

Figure 1. Silhouette scores across k = 2 to k = 9. The red dashed line marks the selected value of k = 6 (score: 0.2595). Scores plateau beyond k = 6, indicating diminishing returns from additional clusters.

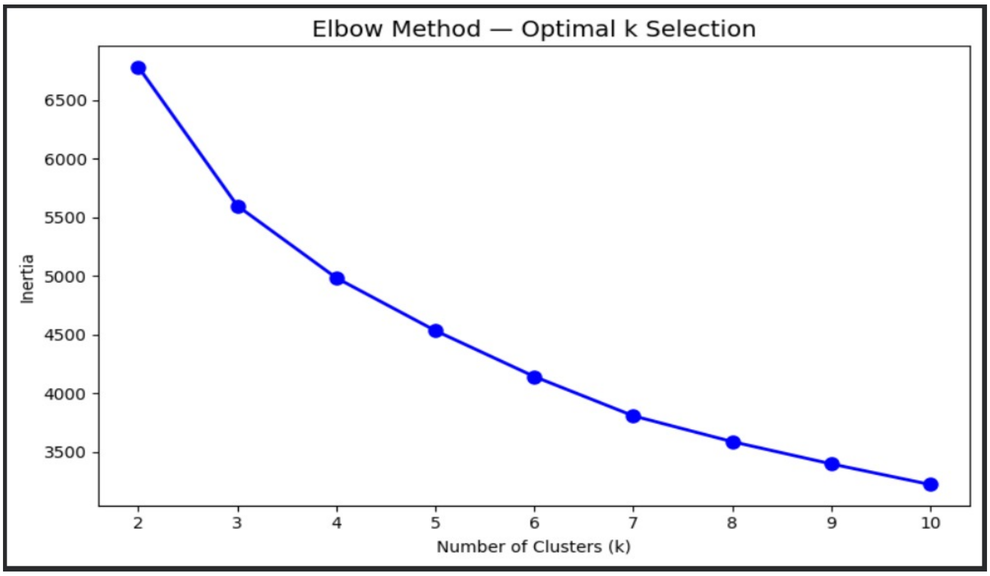

Figure 2. Elbow Method plot showing within-cluster inertia as a function of k. The curve begins flattening noticeably at k = 5-6, consistent with silhouette analysis.

4.2 Cluster Structure

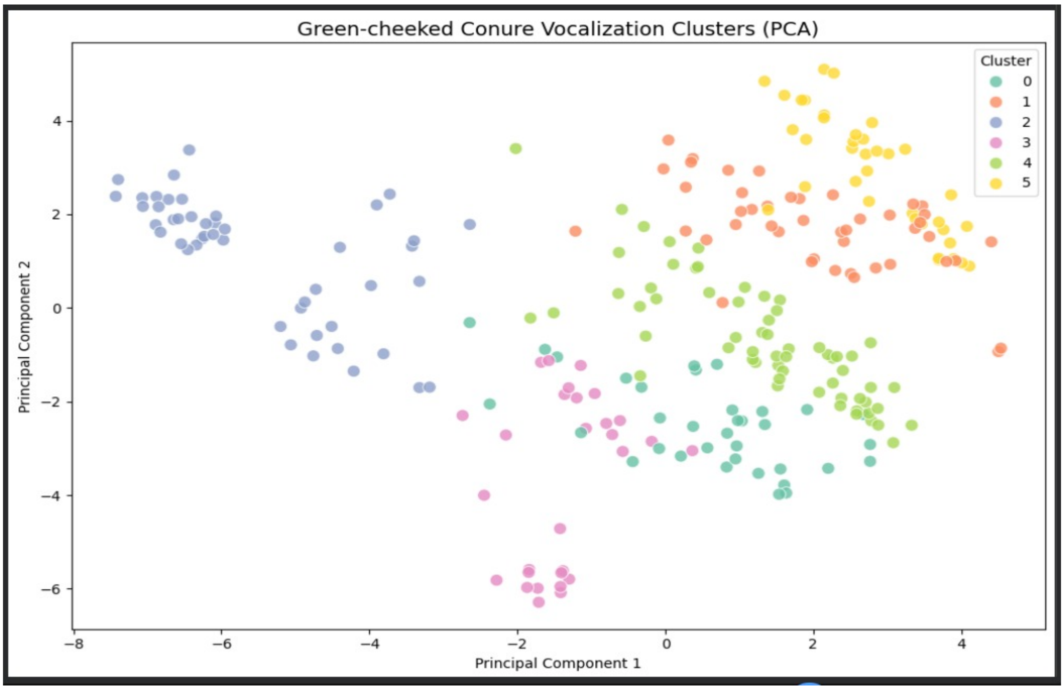

Figure 3. PCA projection of all 254 segments colored by cluster assignment (k = 6). Cluster 2 (blue) demonstrates clear spatial isolation on the left of PC1, indicating a structurally distinct vocalization type. Clusters 0, 1, 4, and 5 occupy overlapping territory, consistent with a continuum of related call types.

The PCA plot reveals a notable finding: Cluster 2 is spatially isolated from all other clusters along Principal Component 1. This separation was consistent across multiple random initializations of K-Means, indicating stability rather than algorithmic artifact. Qualitative listening to segments from this cluster confirmed a distinctive acoustic character — sharper, more tonally focused calls compared to other clusters. This may correspond to vocalizations produced in low-social-density contexts, though behavioral confirmation would require annotated observational data.

4.3 Spectrogram Analysis

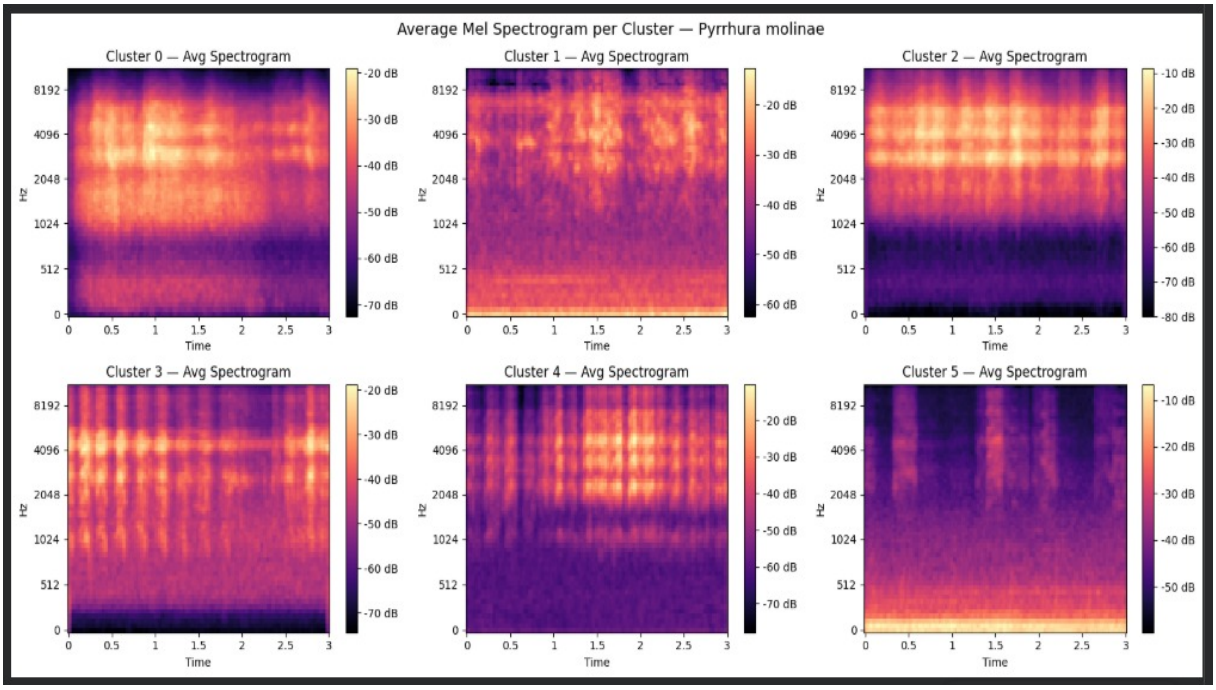

Figure 4. Average Mel spectrograms for each of the six clusters. Distinct energy distribution patterns are visible across clusters, confirming that clustering captured genuine acoustic differences rather than noise.

Average Mel spectrograms per cluster reveal qualitatively distinct acoustic profiles:

- Cluster 1: Concentrated high-frequency energy (above 4096 Hz), suggesting short, sharp contact or alarm calls.
- Cluster 3: Strong horizontal harmonic banding across the full time window, indicating sustained tonal vocalizations with stable pitch.
- Cluster 4: Rhythmic vertical energy banding, consistent with repetitive flock chatter or flight call sequences.
- Cluster 5: Diffuse, low-amplitude energy spread across frequencies, potentially representing background or distant vocalizations.
- Clusters 0 and 2: Moderate, structured mid-range energy, likely representing common contact or social calls.

5. Discussion

The moderate silhouette scores obtained in this study (0.22-0.26 across tested k values) are consistent with findings in comparable bioacoustics research. Animal vocalizations exist on a continuous acoustic spectrum rather than in perfectly separable discrete categories, and unsupervised methods applied to naturalistic recordings routinely produce silhouette scores in this range. The overlapping structure of clusters 0, 1, 4, and 5 in PCA space reflects this biological reality and should be interpreted as a finding in itself: P. molinae vocalizations appear to form a graded acoustic continuum with one or more structurally distinct outlier types.

The consistent isolation of Cluster 2 across analyses is the most significant finding of this study. Its separation along the primary axis of acoustic variance (PC1) suggests that whatever communicative or contextual function it serves, it is acoustically unlike the rest of the repertoire sampled. Future work should prioritize behavioral annotation of recordings containing this cluster's segments to test whether they correspond to a specific social context, arousal state, or interactional function.

A key methodological finding emerged during dataset construction: recording quality significantly impacts cluster stability. Including lower-quality recordings (Xeno-canto grades B-E) substantially degraded silhouette scores, while restricting to grade-A recordings restored clustering consistency. This finding has practical implications for future bioacoustics studies using citizen science databases.

6. Future Work

This project establishes a foundation for a structured research programme in P. molinae bioacoustics. Immediate next steps include:

- Behavioral annotation: Pairing acoustic recordings with video-based behavioral observation to test whether clusters correspond to identifiable behavioral contexts (foraging, alarm, social bonding, flight).
- Expanded dataset: Incorporating additional high-quality recordings to improve cluster stability and enable supervised classification once behavioral labels are established.
- Deep learning approaches: Applying convolutional neural networks (CNNs) to raw Mel spectrograms, bypassing handcrafted feature extraction and potentially revealing finer acoustic structure.
- Cross-individual analysis: Investigating whether vocalization clusters are consistent across individual birds or vary with individual identity, flock membership, or geographic origin.

The long-term aspiration of this work is to contribute to a validated behavioral-acoustic annotation framework for Pyrrhura molinae — the kind of systematic mapping that has begun to illuminate communication in dolphins and great apes, and that represents one of the most compelling open problems at the intersection of biology, linguistics, and artificial intelligence.

7. Technical Summary

Component	Detail
Language	Python 3 (Google Colab)
Audio Processing	librosa, soundfile
Machine Learning	scikit-learn (KMeans, PCA, StandardScaler, silhouette_score)
Data Handling	pandas, numpy
Visualization	matplotlib, seaborn

Data Source	Xeno-canto (xeno-canto.org) — Grade A recordings only
Dataset Size	27 recordings, 254 segments (3.0s each)
Feature Vector	34 dimensions (20 MFCCs + 12 Chroma + Centroid + Contrast)
Algorithm	K-Means Clustering (k=6, n_init=10, random_state=42)
Validation	Silhouette Analysis + Elbow Method
Final Score	Silhouette = 0.2595 at k=6

Data Source: Xeno-canto Bird Sound Database (xeno-canto.org) — recordings used under open access terms for research purposes.